

RESULTS

Results

01 · MANN-WHITNEY U

nonparametric two-group comparison

Table 1. Rank summary

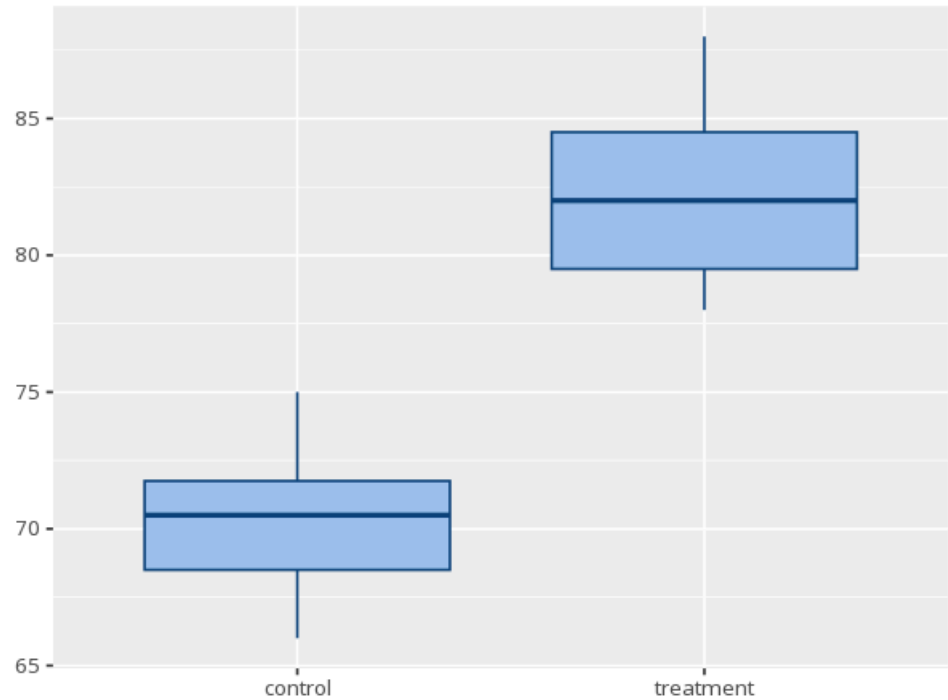
Group	N	Mean rank	Median	IQR	Sum of ranks
control	6	3.50	70.50	3.25	21.00
treatment	6	9.50	82.00	5.00	57.00

Table 2. Mann-Whitney test

U	Z	p	r [95% CI]
0	-2.88	.002	-1.00 [-1.00, -1.00]
Hodges-Lehmann median difference = -12.00, 95% CI [-17.00, -7.00]			

r is the rank-biserial effect size. The Hodges-Lehmann estimate is the median of all between-group score differences (a location shift); its CI is from the same wilcox.test.

Figure. Distribution by group



Understanding the numbers

IQR. The middle 50% spread of scores within each group (descriptive, not part of the test itself). Here, the IQR is 3.25 for control and 5.00 for treatment.

Mean rank. The average rank of each group's scores when all scores are ranked together – the quantity the test compares. Here, the mean rank is 3.50 for control and 9.50 for treatment.

Median. The middle value of each group's scores. Here, the median is 70.50 for control and 82.00 for treatment.

N. The number of observations in each group. Here, $N = 6$ for control and 6 for treatment.

p. The probability of a rank difference this large (or larger) if the two groups truly had the same distribution. Here, $p = .002$ - below the conventional .05 threshold, one group's values are systematically higher or lower.

r. The rank-biserial effect size for the difference between the two groups. Here, $r = -1.00 [-1.00, -1.00]$.

Sum of ranks. The total of the ranks assigned to each group's scores – the quantity U is calculated from. Here, the sum of ranks is 21.00 for control and 57.00 for treatment.

U. The Mann-Whitney test statistic, based on how the two groups' ranks compare. Here, $U = 0$.

Z. The standardized test statistic used to compute p . Here, $Z = -2.88$.

Hodges-Lehmann estimate. The median of every possible between-group score difference — a robust estimate of the location shift between the two groups. Here, the Hodges-Lehmann estimate is -12.00 , 95% CI $[-17.00, -7.00]$.

How to read this test

The nonparametric counterpart to the independent t -test, comparing whole distributions via ranks rather than means. A p below α means one group's values tend to be systematically higher or lower (stochastic dominance) — this equals a difference in medians only when the two groups have similar distribution shapes. r gives the effect size. Your significance threshold (α) is 0.05. Method: Wilcoxon rank sum exact test.

APA template: A Mann-Whitney U test gave $U=0$, $Z=-2.88$, $p = .002$, $r=-1.00 [-1.00, -1.00]$.

Statistical basis: Mann-Whitney U test (Mann, H. B., Whitney, D. R. (1947). "On a test of whether one of two random variables is stochastically larger than the other." *Annals of Mathematical Statistics*, 18(1), 50-60.); Rank-biserial effect size r (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." *Journal of Open Source Software*, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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